

Web Data

CMPSC 105 – Data Exploration



ALLEGHENY COLLEGE

Goals

- Website structure
- Gathering content off the web
- Gathering data about web usage

Website structure

structured text

Raw File Types

- HTML - Hypertext Markup Language
- CSS - Cascading Style Sheets
- XML - Extensible Markup Language
- (also javascript for coding interactions with buttons, and dynamically loading content)

Examples

HTML

- notice tags `<p></p>`, `<div></div>`
- content is inside of tags
- not WYSIWYG like MS Word documents

CSS

- notice metadata marking things like font names and colors
- “font-family:”
- “color:”

Examples

XML

- hierarchical (nested structure)
- tagged metadata
- (javascript)

```
<receipt>
  <store>Walmart</store>
  <items>
    <item>
      <name>Bananas</name>
      <price>0.20</price>
    </item>
  </items>
</receipt>
```

Metadata

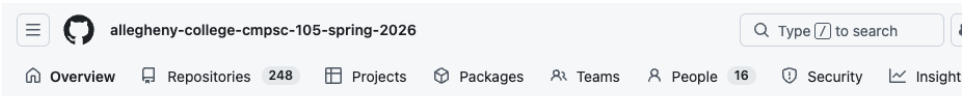
What is Metadata?

- Definition: Metadata is data that provides information about other data.
- Importance: Metadata helps in organizing, finding, and understanding data.
- Examples: Author, date created, file size, etc.

Applications like your browser need to know metadata about content.

- what kind of content is it?
- if it is a paragraph, it should be rendered one way
- if it is an image, it should be rendered a different way
- if it is a url, it should be clickable

Examples



allegheeny-college-cmpsc-105-spring-2026

5 followers Part of Allegheny College

New

Scan for secret leaks with GitHub

Run a free secret risk assessment to determine if you have any exposed secrets in your organization's repositories.

Scan now

Remind me later

Secrets found

Public leaks

Scan sources

Secret types scanned

Token type

Distinct repositories

```
<meta name="user-login" content="hangzhaogogogo">
<link rel="sudo-modal" href="/sessions/sudo_modal">
<meta name="viewport" content="width=device-width">

<meta name="description" content="allegheeny-college-cmpsc-105-spring-2026 has 12 repositories available. Follow their code on GitHub.">
<link rel="search" type="application/opensearchdescription+xml" href="/opensearch.xml" title="GitHub">
<link rel="fluid-icon" href="https://github.com/fluidicon.png" title="GitHub">
<meta property="fb:app_id" content="1401488693436528">
<meta name="apple-itunes-app" content="app-id=1477376905, app-argument=https://github.com/allegheeny-college-cmpsc-105-spring-2026" />
```


Other Markup

- LaTeX - Compiled Markup, often used for rendering
- mathematical expressions
- markdown

markdown

What is Markdown?

Markdown is a lightweight markup language designed for simplicity, allowing users to format plain text with easy-to-read syntax, such as using # for headings and ** for bold text (e.g., **bold**).

```
1 ## Report for Data Exploration
2
3 ### Introduction
4
5 Data was explored using different chart types.
6
7 ### Methods
8
9 Matplotlib functions for making line charts, boxplots, and histograms we
```

Rendered version

Report for Data Exploration

INTRODUCTION

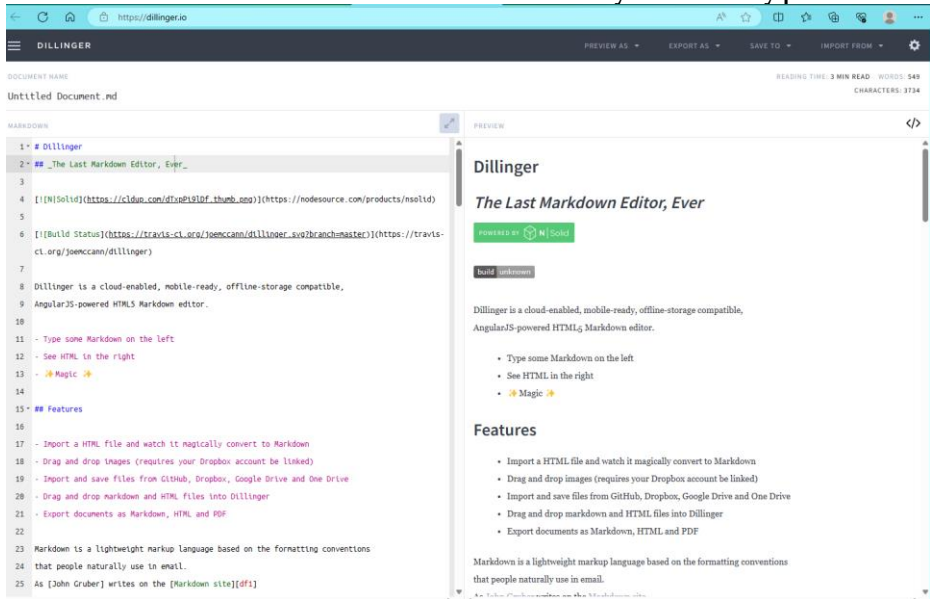
Data was explored using different chart types.

METHODS

Matplotlib functions for making line charts, boxplots, and histograms were used.

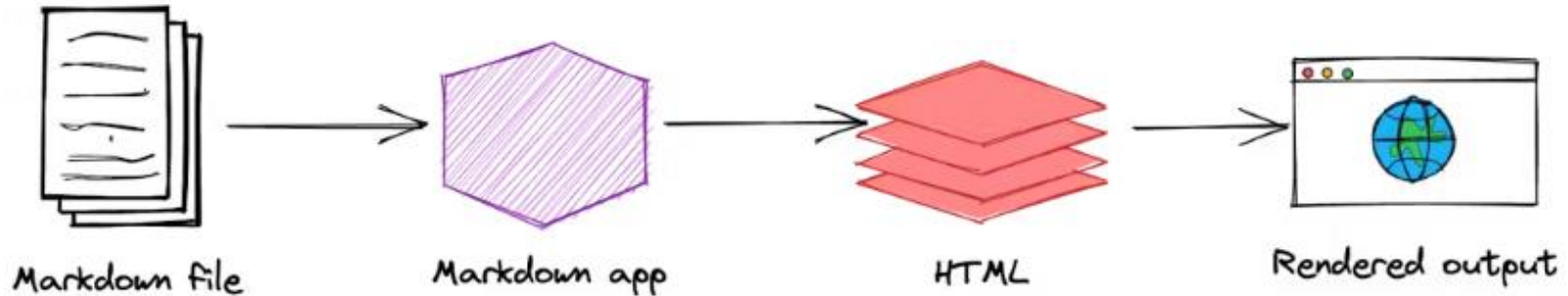
Getting Started with Markdown

To start using Markdown, try it online without downloading anything. For example, use Dillinger (<https://dillinger.io/>), a free online Markdown editor where you can type on the left side and see a preview on the right.



How Does it Work?

- Markdown text is saved in files with an .md or .markdown extension.
- A Markdown application processes the file, converting the Markdown-formatted text into HTML, suitable for web browsers or print-ready documents.
- Markdown applications use something called a Markdown processor (also commonly referred to as a “parser” or an “implementation”) to take the Markdown-formatted text and output it to HTML format or convert it to another file format, like PDF.



Basic Syntax – Code

To denote a word or phrase as code, enclose it in backticks (`).

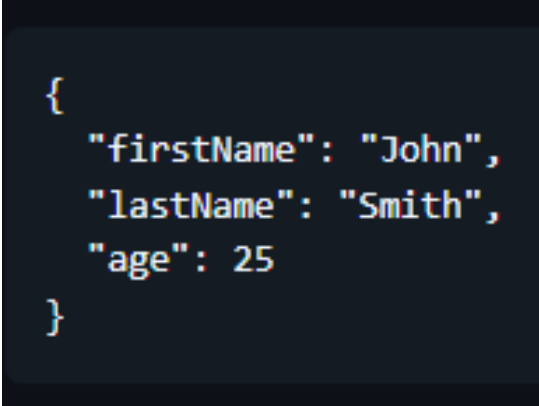
Markdown	HTML	Rendered Output
At the command prompt, type <code>`nano`</code> .	At the command prompt, type <code>nano</code>.	At the command prompt, type nano.

Fenced Code Blocks

Depending on your Markdown processor or editor, you'll use three backticks (```) or three tildes (~~~) on the lines before and after the code block.

```
~~~  
{  
  "firstName": "John",  
  "lastName": "Smith",  
  "age": 25  
}  
~~~
```

The rendered output looks like this:



```
{  
  "firstName": "John",  
  "lastName": "Smith",  
  "age": 25  
}
```

Syntax Highlighting

```
```json
```

```
{
 "firstName": "John",
 "lastName": "Smith",
 "age": 25
}
```
```



```
{  
  "firstName": "John",  
  "lastName": "Smith",  
  "age": 25  
}
```


Basic Syntax – Links

To create a link, enclose the link text in brackets (e.g., [Google]) and then follow it immediately with the URL in parentheses (e.g., (https://www.google.com)).

My favorite search engine is [Google](https://www.google.com/).

The rendered output looks like this:

My favorite search engine is [Google](https://www.google.com/).

Basic Syntax – Images

To add an image, add an exclamation mark (!), followed by alt text in brackets, and the path or URL to the image asset in parentheses. You can optionally add a title in quotation marks after the path or URL.

`![text](URL)`

`![DataExploration](https://github.com/allegHENY-college-cmpsc-105-spring-2026/classDocs/blob/main/graphics/dataE.png)`

Tables

To add a table, use **three** or more hyphens (---) to create each column's header, and use pipes (|) to separate each column. For compatibility, you should also add a pipe on either end of the row.

```
Syntax	Description
Header	Title
Paragraph	Text
```

Cell widths can vary, as shown below. The rendered output will look the same.

The rendered output looks like this:

| Syntax | Description |
|-----------|-------------|
| Header | Title |
| Paragraph | Text |

Website structure

- Websites use structured texts that can be rendered
- Metadata tags are everywhere
- Metadata tags describe the data and can be used to locate data

Gathering Content Off the Web

Gathering content off the web

- discovery
- crawling
- indexing
- scraping

How Things End Up in Search Engine Results

search engine bots, aka crawlers or spiders, need to know about all the pages on a website and all the content on the pages in order to make it indexable, i.e. usable for web searching.

A Google bot has to discover webpages, crawl them, index them, and query a giant database to return related content to search terms.

Discovery

- done with robots.txt
- sitemaps

robots.txt



 YouTube

Search

The Role of robots.txt Files

LinkedIn Learning

SEO Tutorial - The role of robots.txt files

<https://www.youtube.com/watch?v=AyPAfzcVn2c>

robots.txt

← → ↺ 🌐 youtube.com/robots.txt

```
# robots.txt file for YouTube
# Created in the distant future (the year 2000) after
# the robotic uprising of the mid 90's which wiped out all humans.
```

```
User-agent: Mediapartners-Google*
Disallow:
```

```
User-agent: *
Disallow: /api/
Disallow: /comment
Disallow: /feeds/videos.xml
Disallow: /file_download
Disallow: /get_video
Disallow: /get_video_info
Disallow: /get_midroll_info
Disallow: /live_chat
Disallow: /login
Disallow: /qr
Disallow: /results
Disallow: /signup
Disallow: /t/terms
Disallow: /timedtext_video
Disallow: /verify_age
Disallow: /watch_ajax
Disallow: /watch_fragments_ajax
Disallow: /watch_popup
Disallow: /watch_queue_ajax
Disallow: /youtubei/
```

```
Sitemap: https://www.youtube.com/sitemaps/sitemap.xml
```

```
Sitemap: https://www.youtube.com/product/sitemap.xml
```

robots.txt

- directives tell bots where to go / not go (“disallow”)
- helps protect information (in theory) on admin pages or other private pages
- helps keep search engine results relevant
- could prevent duplication if duplicate information is on a website
- includes links to sitemaps!

Sitemaps

- lists out all pages (unique urls) of a website that should be indexed
- these can be used to scrape information off a website
- large websites may have multiple site maps

Sitemaps



XML Sitemap

Generated by **Yoast SEO**, this is an XML Sitemap, meant for consumption by search engines.

You can find more information about XML sitemaps on sitemaps.org.

This XML Sitemap Index file contains 21 sitemaps.

| Sitemap | Last Modified |
|---|-------------------------|
| https://www.whitehouse.gov/post-sitemap.xml | 2026-02-21 05:40 +00:00 |
| https://www.whitehouse.gov/post-sitemap2.xml | 2026-02-21 05:40 +00:00 |
| https://www.whitehouse.gov/page-sitemap.xml | 2026-02-20 13:39 +00:00 |
| https://www.whitehouse.gov/past_event-sitemap.xml | 2026-02-21 19:03 +00:00 |
| https://www.whitehouse.gov/past_event-sitemap2.xml | 2026-02-21 19:03 +00:00 |
| https://www.whitehouse.gov/eop_cea-sitemap.xml | 2026-02-17 20:27 +00:00 |
| https://www.whitehouse.gov/eop_ceq-sitemap.xml | 2026-02-17 16:29 +00:00 |
| https://www.whitehouse.gov/eop_omb-sitemap.xml | 2026-02-13 22:00 +00:00 |
| https://www.whitehouse.gov/eop_ondcp-sitemap.xml | 2026-02-20 15:24 +00:00 |
| https://www.whitehouse.gov/eop_ostp-sitemap.xml | 2026-02-18 20:47 +00:00 |
| https://www.whitehouse.gov/eop_ncd-sitemap.xml | 2026-02-11 20:10 +00:00 |
| https://www.whitehouse.gov/gallery-sitemap.xml | 2026-02-19 18:06 +00:00 |
| https://www.whitehouse.gov/issue-sitemap.xml | 2026-02-05 15:53 +00:00 |
| https://www.whitehouse.gov/media_bias-sitemap.xml | 2026-01-17 15:48 +00:00 |
| https://www.whitehouse.gov/category-sitemap.xml | 2026-02-21 05:40 +00:00 |
| https://www.whitehouse.gov/issue_term-sitemap.xml | 2026-02-20 13:35 +00:00 |
| https://www.whitehouse.gov/media_bias_placement-sitemap.xml | 2026-01-17 15:48 +00:00 |
| https://www.whitehouse.gov/media_bias_reporter-sitemap.xml | 2025-12-11 22:07 +00:00 |
| https://www.whitehouse.gov/media_bias_category-sitemap.xml | 2026-01-17 15:48 +00:00 |
| https://www.whitehouse.gov/media_bias_publication-sitemap.xml | 2026-01-17 15:48 +00:00 |
| https://www.whitehouse.gov/wire_edition-sitemap.xml | 2026-02-20 15:40 +00:00 |

Sitemaps

← → ↺ 🌐 walmart.com/help/sitemap_gm.xml

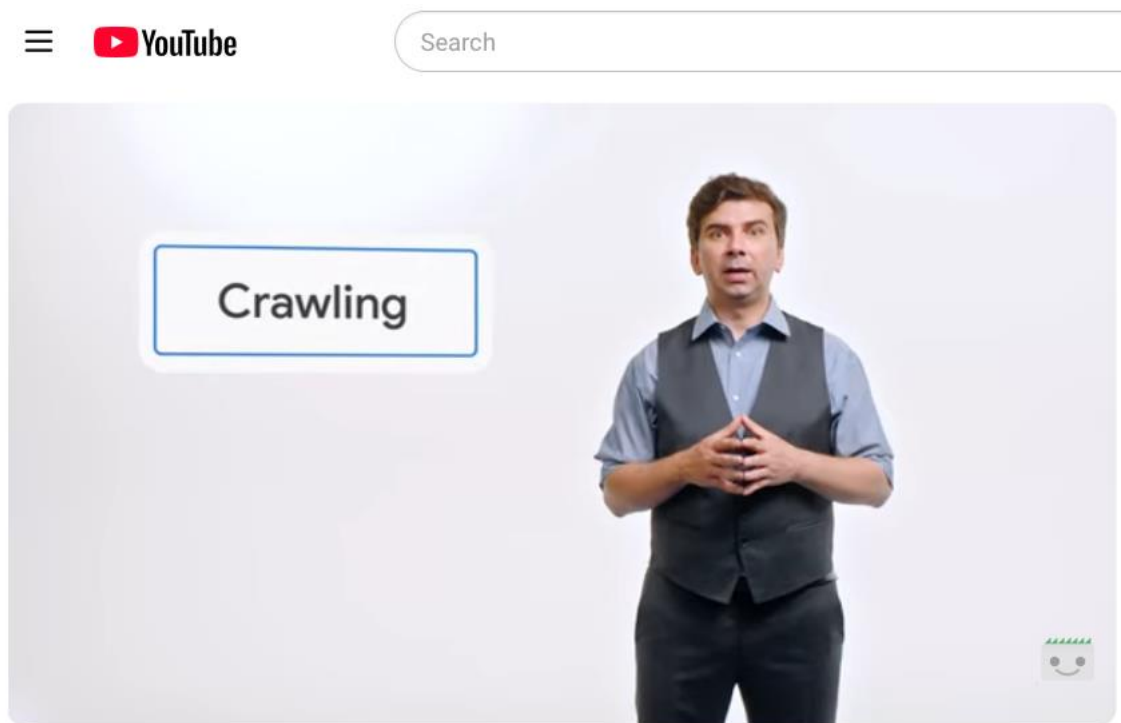
This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  ▼<url>
    <loc>https://www.walmart.com/help/article/track-your-order/143cf6e1d8cb48e6a1ed840409881235</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/edit-or-cancel-an-order/2c7e3d99812044f8aaa65aee0eae3d1c</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/substitutions-for-store-pickup-and-delivery-items/c8dd39735091</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/canceled-orders/8f8dc075d900475a990d4dede6215f0f</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/delayed-orders/1a0e2c333442429ea5b85ccf3c5e408e</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/missing-items/c161a53cd3b7437eb2397d7f0699e0cd</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/order-not-received/af24f9d61b5143d9973f95c9d5bc3140</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
  ▼<url>
    <loc>https://www.walmart.com/help/article/reorder/8c2a6854a3e9428faa93836347c46dd2</loc>
    <lastmod>2026-02-22T01:40:31.448Z</lastmod>
  </url>
</urlset>
```

Participation

- how many sitemaps does Facebook have?
- how many sitemaps does Allegheny have?
- Walmart?

Crawling



How Google Search crawls pages

<https://www.youtube.com/watch?v=JuK7NnfyEuc>

Indexing



How Google Search indexes pages

<https://www.youtube.com/watch?v=pe-NSvBTg2o>

Scraping

- regular people (not Google) can also get content off the web by using strategies similar to discovery, crawling, and indexing.



What is Web Scraping and What is it Used For? | Definition and Examples

<https://www.youtube.com/watch?v=Ct8Gxo8StBU>

Scrapping tools

Code: Requests, BeautifulSoup
Applications:

<https://www.octoparse.com/blog/9-free-web-scrappers-that-you-cannot-miss#10-free-web-scrapping-tools-in-2024>



Summary of gathering content off the web

- thanks to structured text on the web, scraping code or programs can gather information and turn it into useable data for exploration.